

This evolution is underpinned by a national push for 'Made in India' electronics, not just as an assembly hub, but as a design and manufacturing powerhouse. Geopolitical shifts and supply chain disruptions have opened opportunities for India to emerge as an alternative to China. However, this also exposes vulnerabilities, especially in component sourcing, R&D infrastructure, and manufacturing maturity.

DFM: Principles and technical foundations

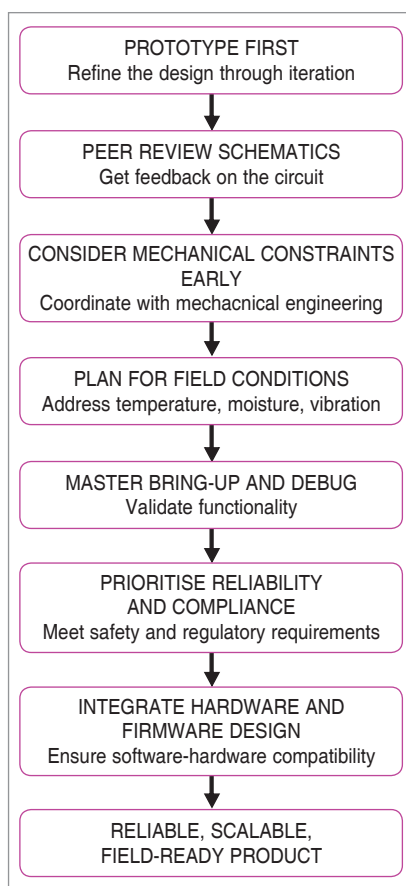
DFM is an engineering approach focused on creating electronic products that are optimised for manufacturing, assembly, testing, and long-term performance. By incorporating DFM principles early in the design process, engineers ensure that products are easy to manufacture at scale, maintain consistent quality, remain cost-effective, comply with relevant industry standards, and perform reliably in real-world conditions.

DFM is central to ensuring that the product moves seamlessly from concept to production. A design should not only be functional at prototype stage but also robust enough to be fabricated, assembled, and tested repeatedly and reliably.

The benefits of DFM are significant across the product lifecycle. By minimising production defects and the need for rework, DFM helps lower overall manufacturing costs. It also facilitates improved automation and scalability, enabling more efficient production processes. Additionally, DFM contributes to enhanced product quality and higher customer satisfaction, while accelerating time-to-market for new products.

India-specific DFM challenges

India remains heavily dependent on imported components—especially semiconductors, sensors, and advanced displays. This exposes the supply chain to cost fluctuations and delays.



Workflow of DFM

Manufacturing infrastructure limitations include:

PCB fabrication. Advanced PCBs (for example, having > 6 layers, fine traces) are challenging to source locally, especially for low-volume runs.

Enclosure fabrication. Quality, low-volume enclosure manufacturing is limited, affecting startups and SMEs.

Testing labs. Access to certified, third-party testing facilities is scarce, causing delays in validation and compliance.

Skills and ecosystem gaps include:

Design expertise. Many engineers lack experience in DFM, compliance, and system reliability.

Academia-industry disconnect. Curricula often lag behind industry needs, especially in applied design and compliance engineering.

Prototyping ecosystem. The ecosystem for rapid prototyping

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